

PIR - Delay in sending sales messages from POS to Sterling, CSSM, BEAM

M&S

Digital&Tech *Post Incident Review*



Incident/ Problem Reference: 90636458/71711

Incident Date: 01/03/2025

PIR Date: 04/03/2025

Author: Saloni Chowdhury

Technical Owner: **William Hanson & Thomas Grigg**

PIR Reviewer: **William Hanson & Thomas Grigg**

1. Incident Summary :

Alerting indicated a delay in sending order payment messages from POS Beanstore to Sterling, CSSM resulting in a cancellation of 121 customer orders incurring a financial loss of £4,872 and inaccurate stock position for ~4997 store UPC combinations across 622 stores leading to stock corruption. Contingency was invoked to generate Foods ordering & allocations with 28 day order plan on hold for 02/03 resulting in Foods analytics reports not being updated with latest order plan information impacting planning activities. The root cause was attributed to the one of the POS channels going into a "retry state" resulting in message backlogs. Services were restored by manually resetting the channel to process the message backlog. Mandated counts were sent to the impacted stores and Foods allocations and ordering were generated using the latest CSSM stock file on 03/03.

2. Incident Chronology :

18:30: MIM bridge initiated.

18:35: There are around 569k sales messages in CSSM (including BP, WHSmith etc). POS is now getting the actual count of transaction with UPC details.

18:44: POS is getting the UPC and store details for the delayed sales messages.

18:46: Unless the impact is ascertained, we are holding the text and comms.

18:50: **As the POS trade queue went into retry queue since 19:12 on 28/02, it is evident that the stock file loaded from CSSM to Relax last night was incorrect - however, very minimal impact as the UK mainchain stores would have been closed by 20:00.**

18:51: Relax has been apprised and the 18:00 Relax Wave 2 batch is cancelled due to inaccurate stock position in CSSM based on Matt's confirmation.

19:26: There is a plan to apply contingency plan for Relax allocations to mitigate impact. However, for correcting store stock positions, a query will be executed to compare the stock value for impacted UPCs and perform manual corrections - takes 2-3 hours.

19:30: Plan to put ASO in backup for Q lines.

19:56: Contingency plan for Foods does not have the user constraints and manual updates made in Relax UI. Therefore, using contingency will be the last option

20:16: **Discussions ongoing with business to proceed with contingency plan for Foods ordering & allocations. Note: The manual recovery is very complex in CSSM and therefore, it was advised to proceed with contingency plan for overnight foods.**

20:18: CSSM team to come back with a recovery plan by considering all possible scenarios by 20:45. There are 66k sales messages potentially impacted, however, we are further drilling down the actual impact considering the below scenarios.

- Store did not receive sales for the day but did counts and corrected the data. No sales after the counting. Sales sent across by evening after counting - need correction with just by the sales values
- Store did not receive sales for the day and did not count and received sales by evening - no need correction
- Store did not receive sales for the day and did counts and it had sale again after counting. Sales was again sent back by evening. - need correction by considering all sales
- Store did not receive sales for the day and did counts after say 6:30 PM - no need correction

21:00: We have identified ~17k UPCs in CSSM which involves the UPCs which the store has counted vs all the messages sent from POS. We are trying to identify the actual count of UPCs impacted by validating the number of UPCs with delayed messages from that one problematic POS node.

21:44: Decision made to invoke contingency for NDC, RDC, Bedworth allocations & SUPPLIER ORDERS.

21:54: Decision made to hold the CSSM security patching scheduled between 01:00 and 04:00.

22:00: We concluded that there should not be a major impact to BEAM reports.

22:54: An OFP banner will be put in for the suppliers to use previous day's 28 day order plan.

23:13: Due to technical challenges, the data could not be exported from SAGG database, therefore, a temp table is being created in database layer and the sales data will be exported to this table for CSSM to validate the actual count of impacted UPCs.

00:42: There are multiple technical challenges in creating the temp table in DB2 layer, we need to identify what can be improved to improve the recovery time.

01:20: After multiple tries, we have now deferred the option of creating temp table and extracting the data from SAGG. Instead we have received the 30 mins delayed sales messages from POS for 1st March. Another dataset for 28th Feb between 19:12 and 00:00 is currently being extracted.

01:23: Impact analysis in CSSM is in progress to identify count of impact store UPC combinations which requires stock correction.

01:39: Recon is in progress on the 17k UPCs by comparing the POS sales feeds to identify the actual count of impacted stores UPC combinations.

01:41: If the impacted UPC count is <60 per store, then we will send mandated counts to stores after agreeing with Retail business

01:45: We have now derived the actual impact - 4997 store UPC combinations across 622 stores with >60 UPCs impacted for 4 stores. 3 out of 4 stores are Franchise stores.

01:46: A decision has been taken to send mandated counts to these 622 stores (including Franchise stores)

01:49: We are reaching out to Alex (Retail business) to agree on sending mandated counts to these stores. In the morning, we will revalidate the UPCs again to exclude the ones for which stores have counted.

2:30 Naga will be speaking to Alex at 06:30 to agree on sending mandated counts to these stores

03:30 Bradford and MK allocation has been generated and sent to WMS by 3:15 and 3:09 respectively

04:00 Relex Final orders have been processed by 03:52, however there is an issue processing FFO in CXM as the outbound processing job is overrunning.

04:50 CXM team has identified a file for the department F02C was missing due to which the job was overrunning. The team has placed a dummy file and has retriggered the CXM job to process the FFO.

06:30 FFO got completed at 06:13

07:07 Naga has spoken to Alex and he requested for Franchise and Main store split to take decision on how to proceed with Mandate count. Out of 662 close 483 stores no action is required.

08:17 List of stores with information of affected UPC's and details if it is a franchise/Main store has been shared to Naga. Naga to speak to Alex and update on the next actions. ETA - 30 minutes

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3. Incident Resolution

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POS - The sender channel within one of the POS Beanstore (there are 4 Beanstore nodes) trade nodes was manually reset, allowing the 520k sales messages to be processed and sent through A9 feed to CSSM, Sterling, Beam and other downstream systems.

CSSM - Within CSSM, the delay in receiving sales messages initially impacted store stock positions. However, the issue was mitigated by identifying the affected store-UPC combinations and, in agreement with the retail business, sending the mandated counts on 3rd March to correct stock positions.

Food - In Foods, the inaccurate stock position within CSSM had a potential impact on RELEX allocations. To mitigate this, the contingency plan was invoked using the previous day's stock file to generate Food ordering & allocations ensuring continued stock distribution to stores despite the delays.

4. Root Cause

Our initial investigation has confirmed that one of the POS channels entered a retry state, leading to a backlog of 520,000 sales messages. However, the underlying cause is still under investigation by the POS and MQ teams. This week, the relevant teams will conduct internal discussions and review logs to pinpoint the root cause of this incident.

Next steps: The logs captured during the incident analysis were shared with IBM, who confirmed that the channel had entered an in-doubt state. As

a result, they have recommended executing specific commands (please check the section at the end of the document) to gather additional information and help determine the root cause. Additionally, work is in progress to automate the channel reset process when it enters a retry state, aiming to prevent downstream impacts. This automation is targeted for completion by 28th March.

Context:

On 28/02, at around 19:12, the POS team received an alert indicating that 2 out of 4 POS BeanStore nodes had entered a retry state. Upon validation, the team confirmed that all channels were restored and functioning as expected. Additionally, they reviewed over 50 internal queues within POS but found no issues. However, a gap in the validation process was identified—the team did not check external channels, which were impacted due to the cascading effect of the retry state, leading to a backlog of 520,000 messages.

During the MQ (Messaging queue) upgrade on the POS gateways on 28/02, an issue was identified, and systemic deployment was completed by 17:00. Pre- and post-checks confirmed that all channels were operational. Log analysis revealed that a communication error occurred on a sender channel of third POS BeanStore node at 19:12, triggering the retry state.

While investigations with the POS and cloud teams continue to determine the root cause, agreed actions include:

- Enhancing alerting within the POS landscape to monitor queue depth per channel.
- Updating the SOP to include validation of external channels.
- Revisiting the alerting scope to trigger additional alerts if queue depth does not return to zero, automatic reset of affected channels to avoid message backlogs.

The key issue here was the lack of visibility into external channels, which prolonged the backlog. Strengthening alerting mechanisms and SOP updates should help prevent similar incidents.

Gap in Impact assessment - When the POS channel was manually reset around 16:30 on 01/03 to process the message backlog, the impact on downstream systems was not fully assessed, leading to stock corruption within CSSM. Moving **forward, if an issue is detected with the POS sales message flow to CSSM, the Q-depth alert will enable immediate action to block stores from counting through the MIM process.**

Sterling/Order Management System: Within Sterling, the team identified orders with missing payment messages since 8 AM. However, as per threshold agreed with .com business, an alert was only triggered once the order count reached 100 at 11:00. To minimize the impact on further orders, the Sterling team adjusted the threshold to 8 hours, ensuring that orders can still be completed even if payment messages are delayed to avoid customer order cancellation.

Reducing the alerting threshold within Sterling increases the risk of false alerts, as valid scenarios exist where customers place orders and pay later, particularly during peak periods. To prevent unnecessary alerts, the threshold for impacted order count was set at 100.

However, a delay in the decision to increase the time threshold between order placement and payment from 3 hours to 8 hours by 14:00 led to an additional 21 order messages being cancelled. The team, following dot-com severity matrix, actively investigated the issue with relevant support teams before making a concurrent decision—approved by the service manager—to increase the threshold.

To mitigate order cancellations, an action has been agreed to assess with the product owners to adjust this time threshold to 6 hours, allowing sufficient time for issue resolution while minimising cancellations.

The process has been revisited to ensure that no approval is required once the order count reaches 100, allowing for a faster response.

Additionally, further discussions will take place with the product owner to reassess the impact threshold before initiating the recovery process. Also, it was highlighted that once the Sterling team received the alert at 100, there was a slow response from the enterprise middleware triage team, further delaying recovery. Actions have been agreed to streamline decision-making, reduce order cancellations, and enhance response efficiency in future incidents.

Refund process: The existing refund process remains with the contact center, who will reach out to customers for refunds or order replacements. Additionally, we are reviewing the refund process for bulk order cancellations to identify a more seamless, system-driven solution that reduces the burden on the contact center and ensures a smoother experience for customers.

CSSM (Central Store Stock Management) : Although it was clear that the message backlog within POS needed to be reprocessed to minimize customer impact, the impact assessment on downstream systems was not conducted efficiently.

If the issue had been identified earlier, store counting could have been blocked to prevent stock corruption. However, since stores counted stock in the morning, the delayed sales messages from POS to CSSM led to discrepancies across 4,997 store-UPC combinations in 622 stores.

Challenges in Impact Assessment within CSSM:

- CSSM does not capture product-level details for sales messages, making it difficult to determine whether a sales message is delayed. While CSSM can detect delayed transactions, it requires store, UPC, and quantity data to assess the full impact.
- The impact assessment process took 5 hours, involving multiple teams (POS, SAGG - Sales Aggregation Middleware, and database teams), which in turn delayed critical decisions related to food ordering and allocations - if the impact would have been identified in a timely manner, the latest stock file would have been used in Relex to generate Food ordering & allocations. **We have therefore agreed that POS will proactively share relevant data (store, UPC, and quantity data) once CSSM provides the list of transactions, ensuring faster assessment.**

In previous instances, the SAGG team was responsible for providing store, UPC, and quantity data to CSSM. The overall process and access within SAG will be reviewed to ensure CSSM receives this information faster, reducing delays in impact assessment.

Alerting

CSSM alerts are configured at the store level (triggered when a store stops sending sales data). However, there is no alert in place to detect overall delays in sales messages, which allowed this issue to go undetected. In this case, only one POS trade node was affected, meaning only a portion of

sales message was delayed, preventing a full alert trigger.

Risk in Manual Recovery

This was a new failure scenario, and CSSM's manual recovery process for delayed sales messages was not well-established. The recovery required a custom query, which had not been tested in lower environments, increasing the risk of execution.

Since the impact was within the cap limit agreed with Retail Ops, a decision was taken to send the mandated counts to stores with agreement with Retail Ops. As a permanent fix, CSSM team will be exploring systemic options to automatically correct corrupted stock data.

5. Findings

5.1. Observability –

During the incident review, we identified gaps across multiple systems—POS, Sterling, and CSSM. In our PIR discussion, we agreed on expanding the scope of alerting within POS, including queue depth alerting, to enable faster issue detection and fine-tune existing alerts to improve accuracy and responsiveness. Also review the Sterling alert threshold with the product owner to reduce customer order cancellation.

Within CSSM, although there is a store level alert for sales messages, it was confirmed that it is technically not feasible to configure an alert specifically for a portion of delayed sales messages from POS to CSSM. Alternative monitoring approaches will need to be considered.

5.2. Technical Challenges -

During the POS recovery process, several challenges were identified that delayed issue resolution including gaps in POS Channel Validation as the initial triage overlooked external channels, delaying issue identification. The SOP is being updated to improve validation processes to ensure both internal and external channels are reviewed during triaging and an option to automate the channel reset process to mitigate the impact.

The major challenges was identified in the impact assessment process due to access issues within SAGG slowed the retrieval of large data volumes, delays in the database layer occurred while creating a temporary table to export the data from SAGG for the CSSM team, therefore, we have agreed that going forward, POS will share relevant data with CSSM to accelerate impact assessment, however, the feasibility is still being explored.

5.3. MIM involvement & impact assessment -

The issue was first highlighted by the .com service manager on an internal channel, who reported a delay in sending order payment messages from POS to Sterling. While the MIM team was involved in the investigation and recovery, the initial impact was classified as SEV-3 under the impact framework, as only 180 customer orders were cancelled over six hours, between 8:00 AM and 2:00 PM. However, once the backlog of 520,000 messages from POS was processed and sent to CSSM, the CSSM team flagged a risk of stock corruption at stores, which would have a cascading effect on food ordering and allocations. At that point, the MIM process was formally invoked, with text, communications and updates shared in the incident channel through loop. The impact assessment was delayed due to various technical challenges (see 5.2), but actions have now been agreed upon to prevent such delays in the future. Moving forward, the focus will be on timely stakeholder notifications, faster impact assessments, and proactive decision-making to minimize business disruption.

5.4. Incident handling/ stakeholder management –

Before and after the MIM process was invoked, the relevant product teams were actively working on the recovery with delays identified in the initial triage process - actions agreed to address the gap. However, the decision to send mandated counts to stores on March 3rd was made solely by Retail Ops, raising concerns about the lack of engagement with the Foods business stakeholders. Ensuring alignment between both portfolios was crucial to maintaining consistency in decision-making. Additionally, concerns were raised about prioritizing business decisions over technical recovery actions without proper concurrence from the Foods business - we have now cascaded the risk of performing manual corrections to the stakeholders. To address this, we have now agreed with the service leads that any business decisions taken to mitigate impact must have joint approval from both Retail Ops and Foods Ops. This will ensure transparency, alignment, and clear visibility across all impacted portfolios.

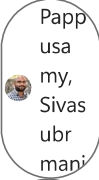
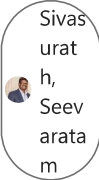
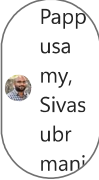
5.5. Business Communication –

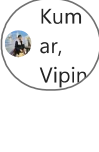
Although there was an initial delay in sending communications due to gaps in impact assessment, once the potential impact was identified, updates were promptly shared with relevant stakeholders. Technical analysis and live updates were continuously provided in the loop, ensuring transparency and keeping stakeholders informed on the actions taken to mitigate the impact and resolve the issue.

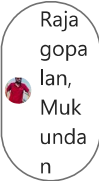
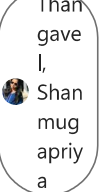
5.6. Supplier Engagement –

While no suppliers were directly involved in the incident recovery, the cascading impact on CSSM leading to stock corruption required engaging the Food RELEX Accenture team. A collective decision was made to implement the contingency plan by highlighting the risks of using an incorrect stock file for food ordering and allocations.

Mitigation Actions:

	≡ Action	👤 Owner	≡ Due date	≡ Remarks/ Updates	📁 Action Category
1	Perform a full analysis of the logs within POS and the MQ environment to determine the root cause of the issue.	 Pappusa my, Sivasubramanian  Bala guru Subramanian	21st March	The logs captured during the incident analysis were shared with IBM, who confirmed that the channel had entered an in-doubt state. As a result, they have recommended executing specific commands to gather additional information and help determine the root cause.	Root Cause
2	5.2 Explore options to gather store, UPC, quantity information from POS in a timely manner to reduce the time taken for impact assessment in CSSM.	 Sivasubrahmaniam  Chelliah, Velrajan	Completed	It has been confirmed that in future scenarios, the relevant information will be provided by the SAGG Team. The requirement to facilitate data extract has been addressed to ensure the information is provided in a timely manner for impact assessment.	Impact Analysis
3	5.4 Review the overall process in SAGG to identify any access issues or technical challenges encountered during the incident in providing the store, UPC, quantity information to CSSM for impact assessment.	 Pappusa my, Sivasubramanian	Completed	During the incident, there was a significant delay from the SAGG team in providing the relevant information to CSSM for impact assessment. To address this, a comprehensive analysis will be conducted to identify the technical challenges or issues faced by the relevant associate. Based on the findings, a plan will be developed to improve the process and ensure timely information is provided, reducing the time required for impact assessment. 11/03: A SOP is created to detail the steps involved including temp table creation in database layer to export the data in SAGG to provide to CSSM. Additional temp table creation is completed in database to be utilised for importing the data to be provided to CSSM for impact analysis	Impact Analysis

4	5.1 Configure a queue depth alert within the POS system to effectively monitor message backlogs in the queues. Additionally, update the Standard Operating Procedure (SOP) to include a step for validating external sender channels during incident triage.	 Bala guru Subra manian	Completed	While there were previous alerts in the POS system, they did not provide sufficient insight for effective issue identification. To address this, a queue depth alert will be configured to detect message backlogs more accurately. The SOP will also be updated to incorporate a step for proactively validating external sender channels, ensuring quicker identification and resolution of potential issues. 11/03: Queue depth alert is implemented in prod.	Problem Identification & Alerting
5	Explore options to automate the reset process for POS channels that enter a retry state.	 Bala guru Subra manian	21st March	During the issue, the impacted POS sender channel required manual intervention for reset, which delayed the recovery process. To mitigate future impact and expedite resolution, we are exploring automation options to reset channels automatically when they enter a retry state. The non prod testing is currently in progress with an ETA to deploy in prod by 28th March	Solution & Recovery
6	5.4 Amend the 3hr time threshold to 6hr within Sterling to reduce the impact on order cancellation count - Agree with M&S POs by assessing any cascading effect before deploying in prod.	 Kumar, Vipin	21st March	The current time threshold in Sterling for customer orders, from placement to payment, is set at three hours. The product team will work with the product owner to extend this to six hours, allowing more time for issue investigations and reducing order cancellations. Implementing this change requires a release process, and a detailed assessment will be conducted to identify any potential cascading effects before deployment to production. 11/03: Work with architects to analyse the cascading effect before increasing time threshold.	Solution & Recovery
7	Revisit the refund process for customer order cancellations.	 Randhawa, Dina  Grigas, Thomas	21st March	Currently, the contact center is responsible for reaching out to customers and processing refunds. However, further discussions between the POS and Sterling teams will take place to explore a more system-driven approach to streamline the refund process.	Solution & Recovery

8	Enhance the incident handling efficiency within Enterprise Middleware team to reduce recovery/issue identification time	 Raja gopalan, Mukunda n	Completed	The team has been asked to review their call rota and designate a technical SME to ensure a quicker and more efficient response to such investigations.	Issue handling/Management
9	Revisit the Sterling time threshold increase process to ensure no approval is required to reduce customer impact	 Kumar, Vipin	Completed	After discussions with the Sterling team, and the dot-com service manager, it has been agreed that no further approval will be necessary to increase the threshold time once an alert is triggered or if there is an impact to 100 customer orders due to delayed payment messages.	Solution & Recovery
10	Perform non prod level testing of the query created within CSSM to manually update stock positions.	 Arjunan, Akash	21st March	Since this was a new scenario in CSSM, the query for manually updating stock positions was created but not tested in a lower environment. To mitigate the risk of further impact, it was agreed with Retail Ops to send the mandated counts to stores on 03/03. The query will now be tested in a lower environment to ensure readiness in case the issue arises.	Solution & Recovery
11	Explore options to systematically correct the corrupted stock in CSSM.	 Arjunan, Akash  Than gave l, Shan mug apriya	21st March	To minimize manual intervention in stock correction, the teams will explore the feasibility of implementing a systemic solution to automatically correct the corrupted stock positions, reducing the risk of manual errors	Solution & Recovery

Please refer to  Major_ Significant and Key Incident Tracker 16th O... for progress updates on Mitigations.

Technical Details and Useful Documents:

Would like to recommend the following tech-note on AMQ9213E :

 <https://www.ibm.com/support/pages/system/files/inline-files/MQ%20errors%20AMQ9219E%20and%20AM...>
<https://www.ibm.com/support/pages/system/files/inline-files/MQ%20errors%20AMQ9219E%20and%20AMQ9213,%20due%20to%20TCP%20error%20ENOTCONN...>

4. Please collect the runmqras data from Queue Manager.

For further troubleshooting, please replace "YourQueueManagerName" with the actual name of your Queue Manager.

```
runmqras -qmlist YourQueueManagerName -section defs -caseno TS018691732
```